15 The nth term of a number sequence is $n^2 - 3$

(a) Find the 4th term of this sequence.

$4^2 - 3 = 16 - 3 = 13$

1 mark
Substitute n = 4.

.....

(1)

(b) Show that 40 is not a number in this sequence.

If $n^2 - 3 = 40$, then $n^2 = 43$.

1 mark
Form an equation.

$6^2 = 36 < 43 < 49 = 7^2$.

So n cannot be a whole number.

1 mark
43 is not a square.

Therefore 40 is not a term.

(2)

(Total for Question 15 is 3 marks)

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